IDETrace split leakage -- what happened and what changes

Internal memo -- David Barron to Dr. Yan Chen -- July 25, 2026

Summary

Our IDETrace evaluation leaked. make_splits.py assigned thirty-second windows to folds with KFold over rows, so windows from the same student were in both training and test folds. With folds grouped by student id, the transformer baseline drops from 0.81 to 0.74 macro-F1. The ranking of models does not change. I propose we report both numbers in the rebuttal (due July 29) and release v1.1 splits grouped by student.

How I found it

While computing per-class F1 for reviewer xT4q, I trained a quick probe to predict student id from the window features as a sanity check. It reached 49% top-1 accuracy over 160 students. That is far above chance (0.6%) and means a model can recognize who wrote a window. I then checked make_splits.py: it shuffles windows, not students.

Corrected results

Model	Random split	Grouped by student	Held-out lab section
Counts + logistic regression	0.69	0.66	0.65
Counts + gradient boosting	0.77	0.71	0.70
GRU	0.79	0.72	0.71
Transformer baseline	0.81	0.74	0.73

What changes

• make_splits.py: GroupKFold(n_splits=5) on student_id, fixed seed; unit test that fails if any student id appears in two folds.

• Datasheet: state the split unit; add the probe accuracy.

• Paper: Section 5 rewritten around grouped results; footnote with original numbers.

• Release: v1.1 splits; v1.0 kept but marked deprecated.

What I need from you

A yes/no on reporting both numbers in the general response (my recommendation: yes, we found it ourselves and it reads better to say so plainly). And whether you want Rezky to re-run the GRU baselines independently before Wednesday.

Related reading

Ferreira and Kowalczyk (JEDM 2023) document the same failure across eleven tasks, with drops of 4 to 17 points; ours (7 points) is in the middle of their range.